

Two Small-Order Classification Theorems for Signed Difference Sets

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Abstract

A signed difference set assigns coefficients in $\{0, +1, -1\}$ to a finite group so that every nonidentity periodic autocorrelation has the same value. We settle two clusters of cases that were marked Open in the April 24, 2026 snapshot of the La Jolla Signed Difference Set Repository. First, a signed $(32, 20, 4)$ difference set exists in an abelian group of order 32 exactly when the group is noncyclic. The six positive cases are represented by explicit coefficient vectors checked by two structurally independent validators; the cyclic case is excluded by a complete $C8 \rightarrow C16 \rightarrow C32$ quotient refinement. Second, no signed $(36, 29, 4)$ difference set exists in any of the three noncyclic abelian groups $C2 \times C18$, $C3 \times C12$, and $C6 \times C6$. Two cases are settled by transparent quotient enumeration. In the last case quotient constraints reduce the problem to one normalized orbit, whose CNF encoding is proved unsatisfiable by a checked DRAT certificate. Together with the cyclic nonexistence already recorded in the frozen repository, this closes the abelian order-36 parameter set. All statements are tied to executable validators, exact hashes, and independently checkable artifacts.

Keywords: signed difference set; finite abelian group; autocorrelation; quotient enumeration; SAT certificate; DRAT.

Main statements

Theorem 1. Let G be an abelian group of order 32. A signed $(32, 20, 4)$ difference set exists in G if and only if G is noncyclic.

Theorem 2. No signed $(36, 29, 4)$ difference set exists in any of $C2 \times C18$, $C3 \times C12$, or $C6 \times C6$.

Corollary. Combining Theorem 2 with the earlier cyclic $C36$ nonexistence recorded in the frozen repository, no abelian group of order 36 admits a signed $(36, 29, 4)$ difference set.

The theorem statements are the focus of this note. The larger 68-entry census from which they emerged is discussed only after their proofs and certificates.

1. Signed difference sets in elementary terms

Let G be a finite group of order v , written multiplicatively with identity e . A signed subset is a group-ring element

$$D = \sum_{g \in G} a_g g, \quad a_g \in \{0, +1, -1\}.$$

Its support has size k when exactly k coefficients are nonzero. Following Gordon [1], D is a signed (v, k, λ) difference set when

$$D D^{(-1)} = (k - \lambda)e + \lambda G. \quad (1)$$

For a reader unfamiliar with group rings, equation (1) is simply a periodic correlation condition. For each shift $h \in G$, form

$$C(h) = \sum_{g \in G} a_g a_{h^{-1}g}.$$

Then $C(e) = k$ automatically, and the defining demand is $C(h) = \lambda$ for every $h \neq e$. One may picture a cyclic group as points on a wrap-around circle and a direct product of cyclic groups as a wrap-around grid. The signs are arranged so that the pattern has exactly the same correlation with each of its nontrivial translates.

Two elementary consequences provide strong arithmetic filters. Applying the trivial character to (1) gives

$$(\sum_g a_g)^2 = k + (v-1)\lambda. \quad (2)$$

For every nontrivial complex character χ , one obtains

$$|\sum_g a_g \chi(g)|^2 = k - \lambda. \quad (3)$$

For $(32, 20, 4)$, equation (2) gives coefficient sum ± 12 . After a global sign change we take the sum to be 12, so every solution has 16 positive and 4 negative coefficients. For $(36, 29, 4)$, the corresponding sum is ± 13 , and the positive normalization has 21 positive and 8 negative coefficients. The nontrivial character magnitudes are respectively 4 and 5.

The invariant-factor notation $[2, 16]$, used by the source repository, means $C_2 \times C_{16}$. The group description is essential: groups having the same order and the same numerical parameters can have different answers, as Theorem 1 shows.

The frozen baseline is commit `e3bf810c5ee6826cf5030f983f6adf23b0ffd20e` of [2], dated April 24, 2026. The target of this note is not a moving web page but the exact named entries in that snapshot.

2. What counts as a computational proof

The distinction between search and proof is central. A heuristic that fails to find a vector says nothing about nonexistence, and a SAT solver printing **UNSAT** is not by itself a durable mathematical certificate. We therefore use different acceptance boundaries for positive and negative results.

For an existence result, the artifact is the full coefficient vector in the repository's lexicographic direct-product coordinate order. A deliberately simple reference validator constructs all group elements as tuples, performs componentwise addition, and recomputes every correlation. A separate validator represents elements by mixed-radix integers and uses different indexing and group-operation code. Both verify the coefficient domain, support, coefficient sum, and all v correlations in unsymmetrized coordinates. A witness is accepted only when both validators agree.

For a nonexistence result, we use one of two routes. A quotient enumeration is acceptable when its finite search space, arithmetic constraints, symmetry normalizations, and final refinements can be reconstructed by a short independent program. A solver-based result must preserve the exact DIMACS formula, the variable mapping implicit in the generator, the DRAT proof, the checker source commit, the checker output, and cryptographic hashes. The proof is accepted only if an independently compiled checker reports **S VERIFIED**.

The certificate chain is therefore:

```
mathematical instance
-> exact reduction or encoding
-> finite enumeration or CNF formula
-> explicit witness or DRAT proof
-> independent full-coordinate validator or proof checker.
```

Symmetry is used only to make a search smaller. Translation, global sign, inversion, and selected automorphisms do not replace the final defining equation. Every positive answer is checked after all normalizations have been forgotten. Every negative symmetry quotient is accompanied by an argument that the action lifts through the next refinement stage.

All accepted files are addressed by SHA-256. The canonical census itself has hash `11cd6bed9b7dcc1c23c66257f0765f2827c8aaefaf944ae4977bf6073cf08d9f`. This makes accidental or selective modification detectable, but hashing is not a substitute for semantic checking: the supplied programs also recompute the mathematics.

3. The quotient projection lemma

The main reduction used in both theorems is elementary. Let K be a normal subgroup of G , let $Q=G/K$, and project D to

$$B = \sum_{q \in Q} b_q q, \quad b_q = \sum_{g \in q} a_g.$$

If $|K|=m$, summing equation (1) over the shifts in each fiber gives

$$\begin{aligned} \text{coefficient of } e_Q \text{ in } B B^{(-1)} &= k + (m-1)\lambda, \\ \text{coefficient of } q \neq e_Q &= m \lambda. \end{aligned} \quad (4)$$

Each b_q is an integer between $-m$ and m , with parity and support restrictions determined by its fiber. Thus a large ternary search is replaced by a much smaller search over bounded integer cell sums. Surviving quotient vectors can then be refined to a quotient with smaller kernel, ending at individual coefficients.

Equation (4) is a necessary condition because projection is a group-ring homomorphism. Completeness requires more: every bounded cell vector compatible with the exact fiber data must be enumerated, and every possible refinement of every survivor must be checked. The independent verification programs do exactly this. They do not trust saved candidate counts; they regenerate the options, correlations, and refinement trees.

Translations act on quotient vectors and lift to translations of the full group. In cyclic quotients, multiplication by a unit also lifts through the cyclic ladder used below. This permits orbit representatives without losing a possible full solution. In product quotients we use only translations unless an additional affine action is explicitly justified.

At order 36, real characters and order-three characters give complementary marginals. For groups with a C_6 factor, the Chinese remainder decomposition $C_6 = C_2 \times C_3$ makes the corresponding translation normalizations independent: the real exceptional fiber and the order-three exceptional fiber can be moved to their chosen origins simultaneously. This observation reduces the final $C_6 \times C_6$ instance to one normalized joint pattern.

The quotient approach has two advantages. It exposes the mathematical reason that the search collapses, and it produces small intermediate data that can be checked without the original solver. Only the last normalized $C_6 \times C_6$ refinement remains large enough to justify a proof-producing SAT encoding.

4. The six constructions at order 32

There are seven abelian groups of order 32. The six noncyclic groups all admit signed $(32, 20, 4)$ difference sets. The table identifies one standalone witness for each group; the final column is the SHA-256 of the JSON witness.

Group	Positive/negative	Witness SHA-256
$C2 \times C16$	16 / 4	4ac02b48513f14385bffd574a2b531d46223375baf7cdc3e398be7f70a433eff
$C4 \times C8$	16 / 4	d22db64868c4632dde42cf06c2f8ad4f36378cec7a8b62f77fa39f022f362540
$C2 \times C2 \times C8$	16 / 4	7b6c903b5b6c7c788615ec9a3ea2b67eb8ff41e538e23085c97b842219144799
$C2 \times C4 \times C4$	16 / 4	b6b9de3b1bd7f821e1d79ffcd03fc92e9931f245429825d1f01e53342e896dc1
$C2 \times C2 \times C2 \times C4$	16 / 4	d099f73f04a3233d8802183e714edfa3cbdc396c252f02dd3dc45dba99910334
$C2^5$	16 / 4	7b4fdd2a34ca1e61376a53169e7475bccedc69aa909d967b5d44011e4eabd101

Each file contains more than a bare vector. It records the group invariant factors, coordinate convention, positive and negative point sets, coefficient sum, complete autocorrelation vector, discovery provenance, and outputs from both validators. The expected correlation vector is

$(20, 4, 4, \dots, 4)$, with 31 occurrences of 4.

The existence proof is short once a vector is known: substitute it into (1). The value of retaining two implementations is defensive rather than theoretical. A shared indexing error can make many vectors appear valid, so the mixed-radix checker deliberately avoids the tuple operation used by the reference checker. Unit tests also compare the implementations on good and mutated vectors.

No classification up to equivalence is claimed. A single verified vector is sufficient for existence in each named group, and the files may represent only one of many translation, sign, inversion, or automorphism orbits. The theorem classifies groups by existence, not the number of inequivalent signed difference sets.

The coefficient composition also gives a convenient human check. Every row has $16+4=20$ nonzero positions and coefficient sum $16-4=12$, consistent with equation (2). This is necessary but not sufficient; the validators still check all 31 nonidentity shifts.

5. Excluding the cyclic group of order 32

For $G=C_{32}$, projection along the subgroup of order 4 first gives a vector on C_8 . The independent exhaustive verifier enumerates 9,528 bounded integral vectors after the total-sum and zero-shift norm constraints. Exactly 56 solve the full quotient autocorrelation equations. The affine action generated by cyclic translation and multiplication by units reduces them to five orbits, of sizes 8, 16, 8, 16, and 8.

Each orbit representative is refined through C_{16} . Across the normalized parents, 12 distinct C_{16} solutions survive. The verifier then enumerates every ternary refinement to C_{32} . There are 20,736 refinements for each of the 12 parents, hence 248,832 final coefficient vectors in total. None satisfies the full C_{32} autocorrelation equations.

Stage	Complete count	Survivors
Bounded vectors on C_8 after sum and norm	9,528	56
Affine orbits on C_8	5	5 representatives
Distinct normalized solutions on C_{16}	all refinements	12
Ternary refinements on C_{32}	248,832	0

The completeness argument has three parts. First, the cell bounds and parity conditions enumerate every possible sum of four coefficients over a C_8 fiber. Second, equation (4) is checked at every quotient shift, not merely at selected characters. Third, every C_{16} fiber is split in all compatible ways, and every surviving two-cell fiber is finally split into its two individual coefficients. The cyclic affine actions used at the first stages lift to C_{16} and C_{32} , so orbit reduction does not discard a possible solution.

The full result is independently regenerated by `scripts/verify_v32_remaining_quotients.py`. The canonical aggregate artifact has SHA-256 2870150476387abada39792d254455a85cd2321a85011c6b2719a756001b1ab. Because all six noncyclic groups have witnesses and the only cyclic group is excluded, Theorem 1 follows.

6. Two transparent exclusions at order 36

For $(36, 29, 4)$, the normalized coefficient sum is 13 and equation (4) combines effectively with real and order-three character marginals.

For $C_2 \times C_{18}$, use a quotient $C_2^2 \times C_3$ with kernel order 3. Its zero-shift correlation must be $29+2 \cdot 4=37$, and every nonzero quotient shift must have correlation $3 \cdot 4=12$. After normalizing the order-three marginal to $(1, 6, 6)$ and the real marginal to $(7, 2, 2, 2)$, the bounded cells lie in $[-3, 3]$. There are 144 candidates satisfying the marginals and norm. Direct calculation of every quotient correlation leaves no solution.

For $C_3 \times C_{12}$, a quotient $C_2 \times C_3^2$ with kernel order 2 has zero-shift correlation $29+4=33$ and nonzero-shift correlation 8. The normalized real marginal is $(4, 9)$, and the order-three marginal is $(-3, 2, 2, 2, 2, 2, 2, 2, 2)$. The cells lie in $[-2, 2]$. Exactly 420 candidates satisfy the marginals and norm, and none satisfies every quotient autocorrelation.

Group	Quotient	Candidates after marginals and norm	Full solutions
$C_2 \times C_{18}$	$C_2^2 \times C_3$	144	0
$C_3 \times C_{12}$	$C_2 \times C_3^2$	420	0

These are exclusions already at a quotient level: a full signed difference set would necessarily project to a listed quotient solution, but there are no such solutions. No SAT solver is used for either decision.

The regenerating checker is `scripts/verify_v36_29_4_combined_quotients.py`. It rebuilds the bounded row and column options, derives the normalized marginals, and checks every shift. The shared aggregate artifact has SHA-256 416ca62582a54b9d3b398436b36ccfbd239db1c48b113ed385329110c98b99e5. This compact file and checker are included in the lightweight package, so these two parts of Theorem 2 can be audited without the gigabyte-scale proof archive.

7. The certified $C_6 \times C_6$ exclusion

The same character analysis for $C_6 \times C_6$ first considers its C_3^2 projection. Exhaustive bounded enumeration produces 106,353 candidates after the sum and norm tests and only nine full projection solutions. Those nine are one affine orbit, represented by

$$(-3, 2, 2, 2, 2, 2, 2, 2).$$

The real quotient likewise has one orbit, represented by $(7, 2, 2, 2)$. Through $C_6 = C_2 \times C_3$, translations can normalize the exceptional cells in both patterns independently. Therefore a single exact full-autocorrelation CNF instance covers every remaining candidate.

The normalized formula has 79,680 variables and 814,764 clauses. The proof-producing solver returned UNSAT and emitted a 23,359,321-byte DRAT trace. The formula and proof hashes are respectively

```
CNF: 3b67af8cc63ea9377220816334fdf26cb1c2c22e5c4dac637f2c9a6f4d3158ff
DRAT: f8a427135933a5033aa33e69554f269c41b3b3544ae00cf2cf18d92b007bb4c4
```

The certificate is checked in forward mode by `drat-trim` at source commit `2e3b2dc0ecf938adbbd779d42877b6ed69d9a985`. The checker validates each admissible proof addition and derives the empty clause; it does not rerun or trust the original solver. A clean-environment audit recompiles that exact checker source in a fresh pinned Debian container, verifies every CNF and DRAT hash, and reruns this proof together with the other accepted proof subcases. The complete machine-readable report and unabridged journal are stored under `artifacts/audit/`.

The mathematical and propositional layers are kept distinct. The quotient enumeration proves that one normalized orbit is complete. The encoding tests compare SAT solutions and direct autocorrelation on smaller instances and known positive cases. The DRAT checker proves that the final CNF has no model. Together these statements exclude a signed difference set in $C_6 \times C_6$ and complete Theorem 2.

The corollary uses one prior datum rather than presenting it as new: the frozen La Jolla snapshot marks $\text{SDS}(36, 29, 4, [36])$ as No with comment `Orbit Exhaust`. The new computations settle precisely the other three abelian groups of order 36.

8. Reproducibility package and audit protocol

The lightweight package is designed to separate ordinary verification from archival storage. It contains this note, all 16 positive witnesses from the broader census, both validators, unit tests, the final quotient artifacts and their regenerating checkers, the census manifest, licensing information, and a SHA-256 manifest. It intentionally omits the 1.1 GB of CNF and DRAT files. Those large traces are to be archived separately with stable hashes and a DOI; the package manifest provides the link once the archival record exists.

A minimal positive audit uses only the Python standard library:

```
python3 -m unittest discover -s tests -v
python3 scripts/audit_final_census.py
```

The three compact final quotient audits are:

```
python3 scripts/verify_v27_remaining_quotients.py
python3 scripts/verify_v32_remaining_quotients.py
python3 scripts/verify_v36_29_4_combined_quotients.py
```

The full certificate audit is intentionally heavier:

```
scripts/run_clean_drat_audit.sh
```

That script pulls a minimal Debian image, records its resolved image digest, installs a compiler and Git inside the disposable container, clones `drat-trim`, checks out the frozen commit, compiles it, and invokes the 57-certificate audit. For each pair the journal records paths, expected and observed SHA-256 values, the exact command, complete checker output, return code, elapsed time, and final verification status. The repository is mounted read-only; the report is copied from a temporary output mount afterward.

Reproducibility has limits worth stating. A DRAT certificate certifies the CNF, not an informal description of the combinatorial problem. Confidence in the translation rests on readable generators, structural tests, agreement with small exhaustive cases, and the quotient completeness argument. Likewise, the dated literature and repository search supports a novelty assessment but cannot logically prove that no earlier result exists.

9. Relation to the 68-entry census, availability, and disclosure

The two theorems are the principal mathematical contribution. They arose in a larger audit of all 68 entries of group order at most 36 marked Open in the frozen snapshot. That census closes every named entry: 16 are existential and 52 are nonexistential. The accepted evidence consists of 16 explicit constructions, 13 decisions by checked SAT/DRAT proofs, 4 direct exhaustive enumerations, 9 exhaustive character obstructions, 14 exhaustive quotient/character reductions, and 12 real-character square obstructions.

A dated prior-art screen found an exact independent predecessor for 58 of the 68 decisions in the public `farev/Matematica` project, with no disagreements. Those 58 decisions are described as independent replications. The seven order-32 and three noncyclic order-36 entries treated in this note are only called novelty-supported: searches through the named repositories, alternate notation, exact parameters, related papers, and citations found no earlier exact resolution as of August 12, 2026. Search evidence is not a proof of novelty, and specialist review remains desirable before journal submission.

All computations reported here used local CPU resources; paid-compute cost was zero. The frozen input data are CC BY 4.0. The project software is released under the MIT License, while the note and project-authored data are released under CC BY 4.0. Exact third-party provenance remains attached to imported material.

AI-use disclosure. OpenAI Codex was used extensively as a computational research and software-engineering assistant: to propose search and quotient strategies, write and revise code and documentation, execute local experiments, organize artifacts, and draft this note. No model output was accepted as mathematical evidence. Every positive claim is backed by an explicit vector checked by two independent deterministic validators; every negative claim is backed by a reconstructible exhaustive calculation or an exact CNF/DRAT certificate checked by independently compiled software. The human author selected the research objective, authorized the work, and remains responsible for the claims, framing, and release.

References

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